



Automated Fish Counting Using Sonar

Issue 2: June 2025

mahobley@caltech.edu

In This Issue...

- [Engineering Development](#)
- [Klamath Deployment](#)
- [Outreach to AI community](#)

Engineering Development

We are designing our pipelines to plug seamlessly into stakeholders' existing workflows, as the goal of this project is to minimize the time and effort required to get accurate counts of salmon populations.

While research into more effective ways to detect, track, and measure the fish is ongoing from the research sub-team, engineering development focuses on this ease of use for the stakeholders.

Since the spring newsletter we have adapted our output interface to interface with Soundmetrics ARIS software, allowing an end user to immediately visualise the results as well as modify them. Currently, we output the detection of each fish as its track crosses the center-line; we have prioritized length measurements as our next task.

Klamath Deployment

The Site of the former Iron Gate Dam



We recognise the need to get Fisheye into the hands of stakeholders early, to be able to implement their feedback effectively. We also recognise that understanding the

complete workflow of the end user – river to data – is important to maximising the utility of our tool.

In early June, some of the Fisheye team travelled to the Klamath River to meet with a non-profit coordinating management of the salmon monitoring, California Trout. We were at their sensor installation, approximately 400m downstream of the Iron Gate Dam site before its removal. We gained a great deal of understanding of the nuances and complications of sensor placement and on-site analysis.

Whilst there, we delivered a pilot version of the Fisheye tool to test it within their data analysis pipeline. We hope to find existing pain-points within our tool for the next few months, iterate, and deliver a set of updates that will let the team use Fisheye during the salmon run later this year. These changes will also improve the tool for future deployment areas.

The Sensor at the Location of the Former Dam



Outreach to AI community

Suzanne, Timm, Justin, Michael, and Sara presented at the Visipedia workshop hosted at MIT. The goal of Visipedia, a collaborative group of researchers at various institutions, is to develop new machine learning techniques and systems for empowering communities of experts. Many of the researchers are focusing on ecological projects, including the apps iNaturalist and Merlin Bird ID. Suzanne, Justin, and Timm presented their research, while Michael and Sara presented on the Fisheye project and the value of this project as well as the value of focusing on developing AI for ecological research.

See other Issues of the newsletter at fish-eye-sonar.github.io

Join the mailing list by emailing: fish-eye-sonar+subscribe@googlegroups.com